Enhancing Robustness in the Wild: Inference-Time Generative Restoration for Occluded Visual Recognition in Affective Computing

Miracle Nnadiukwu [1], Dr. Kazeem Oyebode [1]

[1] Department of Computer and Information Sciences, School of Science and Technology, Pan-Atlantic University, Lagos, Nigeria

Email: miracle.nnadiukwu@pau.edu.ng; koyebode@pau.edu.ng

*Corresponding author

Abstract* - *Deep vision models achieve exceptional accuracy in controlled environments, but suffer severe performance degradation when deployed in-the-wild, where partial obstruction from real-world objects and noise frequently obscure critical data. Conventional robustness strategies rely on training classifiers to tolerate missing information, effectively discarding latent critical signals, or using offline Generative Adversarial Networks (GANs) strictly for offline dataset augmentation. Neither paradigm actively recovers obscure structural cues during live inference.

This study proposes a modular, inference-time generative restoration framework that repairs occluded visual signals prior to inference-time classification, offering a plug-and-play alternative to retraining-heavy robustness pipelines. Using Facial Emotion Recognition (FER) as an empirical benchmark, we evaluate a three-stage progressive pipeline on the RAF-DB dataset. A baseline ResNet-34 trained on clean data achieved 76.4% accuracy on clean inputs but plummeted to 61.4% under occlusion. Standard occlusion-aware retraining improved occluded accuracy to 71.4%, establishing a tolerance-based ceiling. By integrating DeepFill v2 guided by dynamic facial landmark and texture-variance masking as an inference-time preprocessor, the proposed framework restored occluded regions, elevating accuracy to 77.3% and virtually closing the clean-occluded domain gap. These findings demonstrate that active, inference-time signal reconstruction provides a superior pathway for hardening vision classifiers against in-the-wild obstructions without requiring end-to-end retraining.

***Keywords*—affective computing, facial emotion recognition, facial landmarks, generative adversarial networks, image inpainting, occlusion, ResNet-34**

I. Introduction

Deep learning architectures exhibit a generalization bottleneck when transitioning from curated and controlled laboratory benchmarks to unconstrained, real-world environments. Under controlled settings characterized by frontal orientations, uniform lighting, and unobstructed viewpoints, deep convolutional neural networks (CNNs) achieve near-human diagnostic accuracy across visual classification tasks. However, real world vision pipelines deployed "in the wild" encounter unpredictable physical corruptions, most prominently partial occlusion caused by accessories, wearable items, hands, and environmental obstructions. Because deep neural models learn rigid spatial patterns from clean benchmarks, localized visual obstructions distort intermediate feature representations causing massive drops in model reliability and classification confidence.

This vulnerability is particularly pronounced in affective computing, an interdisciplinary domain formalized by Picard [1] to equip intelligent systems with the capacity to interpret and adapt to human emotion states. Within affective computing, Facial Emotion Recognition (FER) serves as an essential perceptual layer for application spanning clinical mental health monitoring, intelligent driver assistance, and interactive pedagogical tools. While deep architectures such as VGGNet, ResNet and mobileNet have replaced handcrafted descriptors (e.g. HOG, LBP) by learning hierarchical features directly from pixel data [2], they remain unreliable under unconstrained conditions. Affective signals are disproportionately concentrated in fine-grained, localized facial structures around the periorbital (eyes, eyebrows) and perioral (mouth) regions. As demonstrated by Zhang et al. [3], standard residual networks degrade when even one of these critical expression-bearing sub-regions is obstructed.

Prior research addressing in-the-wild facial occlusion has predominantly converged on two conventional paradigms:

- Occlusion-Tolerant Classification: This defensive strategy adapts the classifier to operate on corrupted or incomplete inputs through partial-face feature-learning, attention mechanisms that dynamically suppress noisy channels, or feature-masking augmentations during training [4], [5]. Although this paradigm improves classification tolerance, it operates under a fundamental assumption treating occluded regions as irrecoverable noise. Consequently, vital diagnostic information hidden beneath the obstruction such as lip curvature beneath a face mask or brow furrowing behind a hand is permanently discarded rather than recovered.
- Offline Generative Augmentation: A parallel branch that leverages Generative Adversarial Networks (GANs) with contextual attention mechanisms to learn global structural semantics and hallucinate missing pixels [6], [7]. Foundational inpainting frameworks by Yu et al. [6] enable generators to borrow contextual information from visible areas, and coarse-to-fine generative variants have been explored to repair masked facial regions for motion analysis [7]. However, generative methods in visual recognition have been deployed almost exclusively as offline processing tools during training, either to balance skewed class distributions or synthesize occluded variations. During deployment, the live inference pipeline remains passive, forcing classifiers to process corrupted inputs without real-time generative support.

To overcome the limitations of both defensive tolerance and offline augmentation, this paper proposes an active, modular inference-time generative restoration framework. Rather than modifying downstream classifiers to discard missing information or retraining complex backbones from scratch, the proposed model introduces a plug-and-play preprocessing stage that reconstructs occluded features dynamically at runtime prior to feature extraction.

The primary contributions of this work are summarized as follows:

I. Inference-Time Restoration Paradigm: We demonstrate that active, runtime generative reconstruction provides a superior performance ceiling compared to traditional occlusion-tolerant training, recovering latent signals rather than suppressing them.

II. Modular Pipeline Architecture: We design an end-to-end inference framework integrating MediaPipe landmark localization (Lugaresi et al. [15]), paired with patch-level texture-variance analysis, guiding a DeepFill v2 contextual-attention generator (Yu et al. [16]) to restore expression-bearing facial regions in real time.

III. Empirical Validation across Three Evaluative Paradigms: Using an 80:20 partition on the RAF-DB dataset (Li et al. [18]) with an occluded-only test split we evaluate three progressive configurations:

   A. Baseline Transfer Learning: A standard ResNet-34 classifier (He et al. [17]) trained strictly on clean data.

   B. Occlusion-Tolerant Benchmark: An optimized classifier trained on synthetic occlusion pairings and equipped with a channel-attention weighting head, representative of tolerance-based approaches (Chen et al. [4]; Song et al. [11]).

   C. Proposed Generative Restoration Pipeline: An active inference-time system combining MediaPipe perception (Lugaresi et al. [15]) and DeepFill v2 inpainting (Yu et al. [16]) with downstream classification.

The results demonstrate that the proposed restoration framework breaks through the tolerance ceiling achieving a 6.1% accuracy push over the occlusion-tolerant benchmark and virtually eliminating the performance gap between clean and occluded samples.

II. Literature Review

Evolution of Facial Emotion Recognition: From Handcrafted Features to Deep CNNs

Facial Emotion Recognition (FER) has undergone a fundamental paradigm shift from classical feature engineering to end-to-end deep learning pipelines. Early methodologies relied heavily on manually designed texture and geometric descriptors, including Local Binary Patterns (LBP), Histograms of Oriented Gradients (HOG), and Scale-Invariant Feature Transforms (SIFT). While these descriptors, classified via Support Vector Machines (SVMs), performed adequately on early constrained laboratory datasets such as CK+ and JAFFE, they lacked the generalization capacity needed to accommodate in-the-wild variability in illumination, head pose, and facial morphology [8].

The adoption of Convolutional Neural Networks (CNNs) revolutionized visual recognition by enabling models to learn hierarchical spatial representations directly from raw pixel matrices [2]. Foundational architectures such as VGGNet, ResNet (He et al. [17]), and MobileNet have since established baseline performance across standard affective benchmarks. Transfer learning leveraging deep networks pretrained on large-scale datasets (e.g., ImageNet) has become the standard mechanism to accelerate convergence and prevent overfitting on emotion-specific datasets [2].

However, standard CNN architectures exhibit sharp performance degradation when transitioned to unconstrained, real-world deployments. While proprietary or specialized architectures such as NVIDIA's EmotionNet [10] provide competitive benchmark results, they often require rigid execution environments, extensive parameter tuning, or specialized hardware ecosystems, limiting their adaptability as lightweight, plug-and-play solutions. Furthermore, recent explorations with modern architectures such as Vision Transformers [20] and advanced margin-based loss functions [21] emphasize that standard architectural scaling alone cannot fully resolve the distributional shift caused by unpredictable physical corruptions in the wild.

The Occlusion Generalization Bottleneck in Affective Computing

In real-world vision systems, partial facial occlusion, arising from medical masks, sunglasses, winter wear, hands, hair, or severe environmental shadows represents a critical failure mode. Unlike general object recognition where semantic identity can often be inferred from coarse global shapes, affective computing relies on localized, fine-grained micro-expressions concentrated in the periorbital (eyebrows, eyes) and perioral (mouth, lips) zones. When any of these expressive zones are obstructed, deep feature extractors experience severe feature distortion, resulting in misclassifications and collapsed prediction confidence [3], [9]. Zhang et al. [3] documented that standard classifiers routinely experience accuracy degradations exceeding 15% when transitioning from clean to occluded samples.

Prior research addressing in-the-wild facial occlusion has primarily developed along two competing paradigms: *defensive occlusion tolerance* and *offline generative augmentation*.

Defensive Occlusion-Tolerant Strategies

The traditional approach to handling missing facial information focuses on training the classifier to tolerate corruption. One widespread technique is occlusion-aware training, where networks are exposed during optimization to synthetic occlusions such as CutOut, random erasing, or artificial rectangular masks (Song et al. [11]). While synthetic exposure forces the network to discover alternative spatial pathways, the model does not recover the lost diagnostic cues; it merely learns to ignore the occluded patches [11].

A parallel line of work introduces attention mechanisms and partial-face representations to dynamically re-weight feature maps [4], [5]. For example, the Region Attention Network (RAN) proposed by [22] explicitly computes attention weights to prioritize visible facial sub-regions while suppressing occluded channels. Similarly, Ke et al. [5] combined focal transformers with partial feature masking to mitigate missing facial regions. While these tolerance-based strategies improve baseline robustness, they share a fundamental design limitation: they treat occluded regions as irrecoverable noise to be discarded. Crucial diagnostic signals hidden beneath obstructions such as subtle nasolabial folds or lip tremors, are permanently lost to the classification pipeline.

Generative Inpainting: Shifting from Training-Time Augmentation to Runtime Restoration

The second paradigm involves image inpainting using Generative Adversarial Networks (GANs) [12]. Rather than treating corrupted regions as noise, GANs formulate inpainting as a semantic completion problem, hallucinating visually and structurally plausible content conditioned on surrounding uncorrupted context. Foundational generative frameworks, such as Context Encoders by [23], demonstrated that training networks to reconstruct missing regions forces the model to learn deep semantic representations.

Modern inpainting frameworks have significantly advanced structural coherence. Yu et al. [6] introduced contextual attention layers that enable generators to search for and borrow relevant feature patches from distant, unoccluded regions of the image. EdgeConnect [13] further refined this by decoupling structural edge generation from texture synthesis, ensuring sharp boundary reconstruction across complex anatomical contours like the eyes and mouth. In DeepFill v2, Yu et al. [16] replaced standard convolutional operators with learnable gated convolutions, allowing the network to process free-form, arbitrary mask geometries with high fidelity.

Despite the efficacy of generative inpainting, its deployment within affective computing and visual recognition has remained predominantly confined to offline data augmentation. In standard implementations, GANs are utilized prior to training to balance class distributions or synthesize large-scale occluded datasets [14]. Once deployed for live inference, however, the classification model operates without generative assistance, leaving it vulnerable to novel real-world corruptions.

A few recent studies have initiated the transition toward active restoration. Notably, Man and Cho [7] implemented a two-stage coarse-to-fine GAN to repair masked facial regions prior to emotion evaluation. Building upon this trajectory, our work addresses the in-the-wild robustness gap by formulating a decoupled, inference-time restoration pipeline. By pairing dynamic runtime masking derived from MediaPipe 468-point facial mesh tracking [15] and patch-level texture variance with DeepFill v2 contextual inpainting [16], our approach restores occluded affective signals dynamically during inference, providing a modular, plug-and-play enhancement for standard classifiers without requiring end-to-end retraining.

III. Research Methodology

The methodology is structured around a progressive, comparative framework designed to determine if active generative restoration at inference time provides a measurable performance push beyond the ceiling of traditional defensive classifiers. By systematically isolating the transfer learning baseline and the occlusion-tolerant benchmark, any performance gains achieved in the final configuration are strictly attributable to the generative restoration module rather than downstream classifier tuning.

Dataset Preparation and In-the-Wild Partitioning

All experiments were conducted on the Real-World Affective Faces Database [18]. RAF-DB contains diverse facial expressions exhibiting unconstrained real-world variability in lighting, pose, and natural occlusions. To evaluate model resilience under in-the-wild corruptions, the dataset was partitioned using an 80:20 split:

- Training Partition (80%): Composed of clean images and synthetically occluded pairs used to optimize the classification backbones and tolerance baselines.
- Evaluation Partition (20%): Reserved strictly for occluded facial samples. Testing exclusively on obstructed faces provides a direct stress-test of in-the-wild diagnostic performance, avoiding artificial score inflation from clean samples.

Progressive Evaluative Framework

Baseline Transfer Learning Classifier (He et al. [17]):
To quantify the fundamental fragility of unaugmented deep learning models under in-the-wild corruptions, a standard ResNet-34 architecture pretrained on ImageNet was fine-tuned exclusively on clean training images. Because standard residual networks learn rigid spatial representations from clean distributions, this baseline establishes the lower-bound diagnostic degradation when critical facial regions are occluded.

Occlusion-Tolerant Benchmark (Chen et al. [4]; Song et al. [11]; Wang et al. [22]):
To establish the performance ceiling achievable through defensive tolerance-based learning alone, a second ResNet-34 model was trained on a balanced 1:1 mixture of clean and synthetically occluded images. The standard output layer was replaced with a custom dual-layer attention-aware classification head to learn dynamic feature weighting. This head computes channel-wise attention weights (α) to prioritize uncorrupted spatial pathways:

$$\alpha = \sigma(W_2 \bullet ReLU(W_1 \bullet x)) \quad (1)$$

$$x_{attended} = \alpha \odot x \quad (2)$$

$$\hat{y} = W_{cls} \bullet x_{attended} \quad (3)$$

where W_1 and W_2 are learnable projection weights, σ denotes the sigmoid activation function, and $\odot$ represents the Hadamard element-wise product.

To penalize classification errors on obstructed samples, an Occlusion-Aware Cross-Entropy Loss

$$L_{total} = L_{CE} + \beta \bullet L_{penalty} \quad (4)$$

where L_{CE} represents class-weighted cross-entropy, $L_{penalty}$ is the explicit loss penalty for misclassified occluded samples, and β is a scaling hyperparameter.

Proposed Model: GAN-Augmented Inference-Time Restoration Pipeline:

The proposed model transitions from defensive noise suppression to proactive signal restoration. Rather than forcing the classifier to rely on partial faces, DeepFill v2 (Yu et al. [16]) is integrated as a decoupled, plug-and-play preprocessor during live inference.

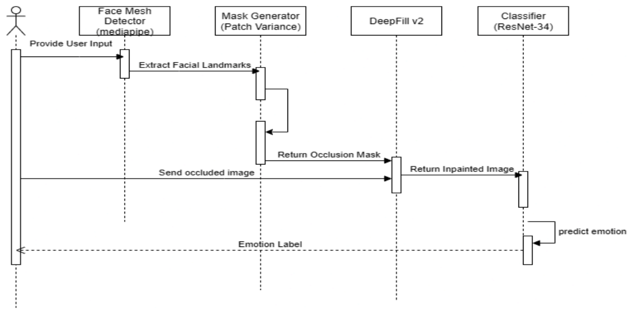

Fig. 1. Proposed Sequence Diagram of the Proposed GAN-Augmented Inference Pipeline.

Dynamic Inference-Time Restoration Logic

Unlike training-time augmentation, the proposed restoration pipeline operates in real time during live deployment when ground-truth occlusion masks are unavailable.
The framework dynamically detects and inpaints corruptions in three sequential steps:

- Step 1: Facial Landmark Localization:The input image is processed through the MediaPipe perception pipeline [15], which extracts 468 high-density 3D facial landmarks to define the facial convex hull and isolate anatomical regions of interest.
- Step 2: Dynamic Texture-Variance Mask Estimation:Within the localized facial boundary, the image is divided into uniform $10 * 10$ pixel patches. Patches exhibiting local pixel variance below a predefined heuristic threshold, characteristic of uniform, flat textures like medical masks, sunglasses, or solid obstructions, are dynamically flagged to generate a binary occlusion mask.

Fig. 2. Landmark-Guided Dynamic Mask Generation. Illustration of the transition from (a) raw input to (b) 468-point facial mesh localization and (c) final patch-based variance mask estimation.

Source: Base image adapted from Pexels (available at:[https://www.pexels.com/photo/close-up-photo-of-a-woman-removing-her-blue-sleep-mask-6963176/] and processed for the experimental pipeline..

- Step 3: Contextual Inpainting and Classification:Guided by the dynamically generated binary mask, the DeepFill v2 generator reconstructs the missing facial cues using gated convolutions and contextual attention layers that copy relevant structural details from uncorrupted facial areas [16]. The restored, complete facial representation is then passed to the downstream classifier for final emotion prediction.

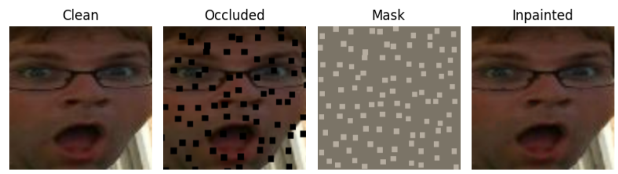

Fig. 3. Generative Image Restoration Pipeline. (a) Input image from the RAF-DB dataset; (b) Synthetic patches added to emulate occlusion; (c) final patch-based variance mask estimation; (d) High-fidelity restoration using the DeepFill v2 contextual attention GAN.

Source: Samples extracted and processed from the Real-World Affective Faces Database (RAF-DB) [19].

Implementation and Optimization Configuration

All classification models were implemented in PyTorch [19] and executed on a GPU-accelerated workstation over 50 training epochs. Training optimization utilized the Adam optimizer coupled with a `ReduceLROnPlateau` scheduler (patience factor of 5 epochs) to maintain stability.

A tiered learning rate optimization strategy was employed to preserve low-level spatial filters while adapting high-level representations to complex occlusions:

- Convolutional Backbone (Layer 4): Assigned a fine-tuning learning rate of $3 * 10^{-4}$ to retain foundational spatial features.
- Attention-Aware Head: Optimized at a learning rate of $1 * 10^{-4}$ to aggressively adapt channel-weighting parameters to occlusion patterns.

IV. Experimental results and discussion

All experimental evaluations were conducted using PyTorch on a GPU-accelerated computing environment. Classification models followed a standardized 50-epoch training schedule, with the validation loss plateauing around epoch 45 across configurations.

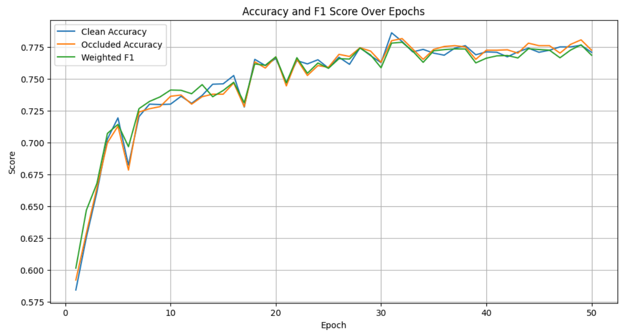

Fig. 4. Performance Convergence of the GAN-Augmented (Proposed Restoration) Pipeline. The plot illustrates the comparative accuracy between clean and GAN-restored occluded samples across 50 training epochs.

Note the convergence of clean and restored accuracy, indicating the effectiveness of the inference-time inpainting in recovering emotional signals.

As illustrated in Fig. 4, the clean and GAN-restored occluded accuracy trajectories converged tightly throughout optimization, maintaining an average difference of less than 1% across epochs. At peak performance, the occluded test accuracy slightly exceeded clean accuracy by a 0.2% margin (77.3% vs. 77.1%). This indicates that inference-time contextual reconstruction not only recovers missing visual structures but can also clarify subtle expression cues that remain ambiguous or noisy in unoccluded in-the-wild input.

Comparative Performance Analysis across Paradigms

To validate the primary thesis, all three evaluative configurations were tested against the 20% occluded test partition of the RAF-DB dataset.

Model Evaluation Paradigm	Occluded Accuracy (%)	F1-Score
Baseline Transfer Learning Classifier (He et al. [17])(Trained on clean data)	61.4	0.56
Occlusion-Tolerant Benchmark (Chen et al. [4]; Wang et al. [22])(Trained on 1:1 clean/occluded mix)	71.4	0.70
Proposed Model: GAN-Augmented Inference-Time Restoration Pipeline	77.3	0.76

Table. 1. Quantitative Performance Metrics on Occluded Test Samples across Experimental Paradigms.

The empirical results in Table 1 demonstrate a clear two-phase performance push: Baseline Fragility to Tolerance Ceiling (+10.0%):

- Exposing the baseline ResNet-34 classifier to occluded test samples resulted in severe degradation, dropping from 76.4% on clean data to 61.4% on occluded data. Training on a 1:1 clean/occluded mix combined with the attention-aware head (Occlusion-Tolerant Benchmark) improved occluded accuracy to 71.4% (F1-score: 0.70). This 10.0% gain establishes the theoretical ceiling of defensive noise-suppression techniques.
- Active Restoration Push (+5.9% / +6.1%): Integrating the DeepFill v2 inpainting module at inference time elevated occluded classification accuracy to 77.3% (F1-score: 0.76). This provides empirical confirmation that active, inference-time generative reconstruction outperforms defensive tolerance strategies by directly restoring diagnostic affective cues rather than treating occlusions as discarded noise.

Per-Class Performance and Error Analysis

To evaluate the class-level behavior of the proposed restoration pipeline, a detailed per-class F1-score and confusion matrix analysis was conducted on the occluded evaluation set.

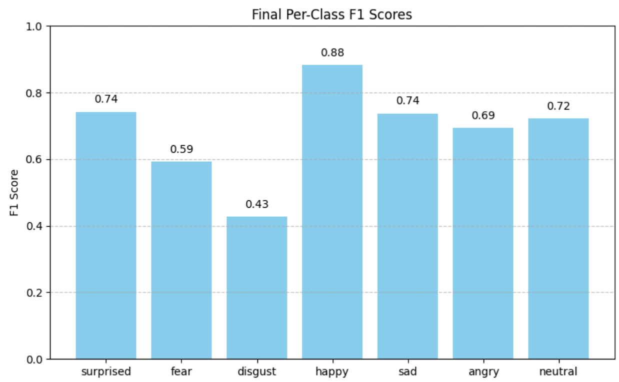

Fig. 5. Detailed F1-Score Performance across Emotion Categories for the Proposed Restoration Pipeline. The bar chart illustrates the model's reliability in classifying specific affective states following GAN-based restoration on occluded samples.

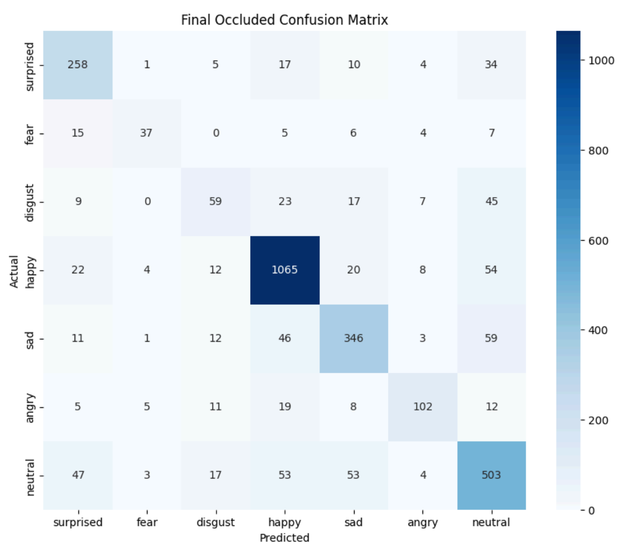

Fig. 6. Confusion Matrix of the GAN-Augmented FER Pipeline on the RAF-DB Occluded Test Set. Diagonal elements represent the precision of predictions for each emotional class, while off-diagonal elements illustrate inter-class misclassification patterns.

As shown in Fig. 5 and Fig. 6, the proposed pipeline achieved superior performance on expressive affective states:

- High-Fidelity Classes: Happiness (0.88 F1) and Surprise (0.74 F1) exhibited the highest classification fidelity. These expressions involve prominent geometric deformations (e.g., broad smiles, raised brows, open mouths) that provide strong structural priors for contextual GAN inpainting. Neutral expressions also demonstrated strong reliability (0.72 F1) due to consistent facial symmetry.
- Challenging Fine-Grained Classes: Subtler emotions, specifically Disgust (0.43 F1) and Fear (0.59 F1), yielded lower recognition rates. This behavior is attributed to two factors: (i) the nuanced, micro-muscle movements characteristic of disgust (e.g., nose wrinkling) are more susceptible to generative smoothing during inpainting, and (ii) the inherent class imbalance of the RAF-DB dataset limits the representation of these categories during backbone training.

Consolidated Reliability Metrics

To ensure that performance gains were not biased toward majority classes in the skewed RAF-DB distribution, macro-level evaluation metrics were aggregated across all emotion categories.

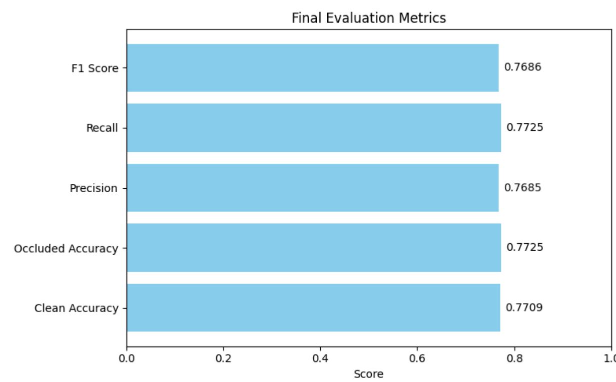

Fig. 7. Consolidated Evaluation Metrics for the GAN-Augmented (Proposed Restoration) Pipeline. The bar chart illustrates the equilibrium between Accuracy, Precision, Recall, and F1-Score, validating the model's reliability on the imbalanced RAF-DB occluded test set.

The consolidated metrics (Accuracy: 0.7725, Recall: 0.7725, Precision: 0.7685, F1-Score: 0.7686) demonstrate balanced equilibrium across predictive dimensions. The tight grouping in the 76–77% band confirms that the proposed modular restoration pipeline hardens the classifier against in-the-wild occlusions by restoring expression-bearing regions prior to downstream inference.

V. CONCLUSION AND FUTURE WORK

This study investigated the integration of Generative Adversarial Networks (GANs) into Facial Emotion Recognition (FER) pipelines to overcome the diagnostic degradation caused by unconstrained facial occlusion in the wild. By transitioning from a defensive paradigm of occlusion tolerance to an active paradigm of inference-time generative restoration, a modular, decoupled framework was developed and empirically validated. The experimental findings confirm that active runtime restoration provides a superior performance ceiling compared to conventional training-time robustness techniques. While the occlusion-tolerant benchmark achieved a 71.4% accuracy ceiling through synthetic data augmentation and attention-aware channel weighting, integrating the DeepFill v2 inpainting engine as a live preprocessor elevated occluded classification accuracy to 77.3% on the RAF-DB benchmark. By dynamically detecting occluded regions via MediaPipe landmark tracking and texture variance analysis, the proposed pipeline effectively recovered missing affective signals and periorbital micro-expressions, allowing downstream classifiers to operate on complete visual representations without requiring end-to-end retraining. Despite the empirical gains achieved by the proposed pipeline, several technical limitations remain to be addressed:

- Semantic Emotion Fidelity: Pretrained DeepFill v2 weights are optimized for generic visual naturalness rather than emotional preservation. Consequently, generative reconstruction may occasionally introduce subtle semantic drift that deviates from the subject's ground-truth emotional state.
- Heuristic Masking Sensitivity: The reliance on patch-level texture variance for dynamic mask estimation can occasionally lead to false positives, misinterpreting environmental shadows or uniform skin regions as physical occlusions.
- Computational Latency: Integrating a generative model into the live inference pipeline introduces additional computational overhead. While viable for real-time execution on GPU-accelerated workstations, further latency optimization is required for constrained edge devices.
- Dataset Distribution Skew: Inherent class imbalances in naturalistic datasets like RAF-DB continue to constrain performance on minor, nuanced affective states such as Fear and Disgust.

To build upon the foundation established in this work, future research should explore the following directions:

- Emotion-Preserving Generative Objectives: Designing domain-specific GAN or diffusion loss functions conditioned on facial action units (AUs) to enforce strict emotional consistency during inpainting.
- Semantic Occlusion Segmentation: Replacing heuristic variance masking with lightweight, learned segmentation networks trained to dynamically detect diverse real-world occlusions with pixel-level precision.
- Advanced Backbone Integration: Evaluating the plug-and-play restoration module with modern Vision Transformer (ViT) and EfficientNet backbones to capture richer global contextual dependencies.
- Edge-Device Optimization: Applying model compression techniques, such as knowledge distillation and structured pruning, to the generative inpainting engine to enable efficient, low-latency deployment in mobile and interactive embedded systems.

Conflict of Interest

The author declares that there is no conflict of interest regarding the publication of this paper.

Author Contributions

MN developed the model's algorithm; MN wrote and documented the manuscript; KO did the final editing and correction; all authors had approved the final version.

Funding

This research was supported by the School of Science and Technology at Pan-Atlantic University through the provision of facilities and materials within the Virtual Reality Lab. No external third-party funding was utilized for this work.

References

[1] R. W. Picard, Affective Computing. Cambridge, MA, USA: MIT Press, 1997.
[2] S. Li and W. Deng, "Deep facial expression recognition: A survey," IEEE Transactions on Affective Computing, vol. 13, no. 3, pp. 1195–1215, Jul.–Sep. 2020.
[3] L. Zhang, B. Verma, D. Tjondronegoro, and V. Chandran, "Facial expression analysis under partial occlusion: A survey," ACM Computing Surveys, vol. 51, no. 2, pp. 1–49, 2018.
[4] Y. Chen, S. Liu, D. Zhao, and W. Ji, "Occlusion facial expression recognition based on feature fusion residual attention network," Frontiers in Neurorobotics, vol. 17, p. 1250706, 2023.
[5] L. Ke, C. Liao, and C. Hsia, "Improving facial expression recognition with a focal transformer and partial feature masking augmentation," Engineering Proceedings, vol. 70, no. 1, p. 70, 2024.
[6] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, "Generative image inpainting with contextual attention," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2018, pp. 5505–5514.
[7] Q. Man and Y. Cho, "Efficient face region occlusion repair based on T-GANs," Electronics, vol. 12, no. 10, p. 2162, 2023.
[8] B. Ko, "A brief review of facial emotion recognition based on visual information," Sensors, vol. 18, no. 2, p. 401, 2018.
[9] V. Chandak, P. Saxena, M. Pattanaik, and G. Kaushal, "Semantic image completion and enhancement using deep learning," in Proc. 13th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT), 2019, pp. 1–6.
[10] NVIDIA, "EmotionNet Model Card," NVIDIA NGC Catalog, 2024. [Online]. Available: https://catalog.ngc.nvidia.com/orgs/nvidia/teams/tao/models/emotionnet
[11] L. Song, M. Zhang, X. Wu, R. He, and T. Tan, "Pairwise differential siamese network for face recognition under

occlusion," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 42, no. 12, pp. 3073–3088, Dec. 2020.
[12] I. Goodfellow et al., "Generative adversarial nets," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2014, pp. 2672–2680.
[13] K. Nazeri, E. Ng, T. Joseph, F. Z. Qureshi, and M. Ebrahimi, "EdgeConnect: Generative image inpainting with adversarial edge learning," in Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW), 2019, pp. 3261–3270.
[14] X. Zhu, Y. Liu, Z. Qin, and J. Li, "Data augmentation in emotion classification using generative adversarial networks," in Proc. Pacific-Rim Conf. Multimedia (PCM), 2017, pp. 574–584.
[15] C. Lugaresi et al., "MediaPipe: A framework for building perception pipelines," arXiv preprint arXiv:1906.08172, 2019.
[16] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, "Free-form image inpainting with gated convolution," in Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV), 2019, pp. 4471–4480.
[17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 770–778.
[18] S. Li, W. Deng, and J. Du, "Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2017, pp. 2584–2593.
[19] A. Paszke et al., "PyTorch: An imperative style, high-performance deep learning library," in Proc. Adv. Neural Inf. Process. Syst. (NeurIPS), 2019, pp. 8024–8035.
[20] F. Zhu, J. Zhu, and W. Dai, "Masked facial expression recognition via vision transformers," IEEE Transactions on Affective Computing, vol. 14, no. 4, pp. 2890–2901, 2023.
[21] K. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR), 2019, pp. 4690–4699.
[22] K. Wang, X. Peng, J. Yang, S. Lu, and Y. Qiao, "Region attention networks for pose and occlusion robust facial expression recognition," IEEE Transactions on Image Processing, vol. 29, pp. 4057–4069, 2020.
[23] D. Pathak, P. Krahenbuhl, J. Donahue, T. Darrell, and A. A. Efros, "Context encoders: Feature learning by inpainting," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR), 2016, pp. 2536–2544.